

Governing Inequality in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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Abstract

The governing inequality of system stability in Dimensional Human Field Theory (DHFT) establishes the formal condition under which structural stability is maintained or violated.

It defines the boundary between stability and instability as a function of load accumulation relative to dissipation capacity.

The inequality is defined by invariant system variables and introduces no additional constructs.

I. Field Context

Dimensional Human Field Theory (DHFT) is a closed explanatory system defined over invariant variables:

- Load (L)
- Capacity (C)
- Boundary (B)

These variables and their structural relationships are non-modifiable.

DHFT is defined within Structural Stability Science (SSS) and derived through Structural Identification Method (SIM).

II. Purpose

The governing inequality defines:

- the formal condition for system stability
- the boundary between stability and instability
- the structural relationship between load accumulation and dissipation

It establishes the canonical inequality governing system state conditions.

III. Governing Inequality

System stability is defined by:

$$dL/dt \leq \kappa(C, B)$$

System instability is defined by:

$$dL/dt > \kappa(C, B)$$

IV. State Correspondence

The governing inequality defines the formal condition underlying system state classification.

Stability corresponds to conditions under which $dL/dt \leq \kappa(C, B)$.

Instability corresponds to conditions under which $dL/dt > \kappa(C, B)$.

Structural load, drift, and failure are classified through evaluation of the relation among L , C , and B over time under the governing inequality.

System state classification is defined in *System State and Failure Conditions in DHFT*.

V. Temporal Evaluation

System state classification is determined over time through evaluation of the relation among L , C , and B .

Instantaneous condition does not alone determine system state.

Sustained violation or satisfaction of the governing inequality defines transition between states.

Temporal evaluation preserves distinction between structural load, drift, and failure.

VI. Variable Definitions

- L (Load): total structural demand
- C (Capacity): finite structural threshold for processing load
- B (Boundary): structural regulator of load entry, containment, and transfer
- $\kappa(C, B)$: dissipation function defined by the interaction of Capacity and Boundary

Boundary modulates $\kappa(C, B)$ through mediation of load flow.

VII. Structural Interpretation

Stability occurs when load dissipation meets or exceeds load accumulation.

Instability occurs when load accumulation exceeds dissipation capacity.

These conditions are independent of:

- domain
- interpretation
- descriptive framing

VIII. Constraint Statement

The governing inequality is defined within the invariant structure of DHFT.

No correspondence in language, form, or method to existing disciplines constitutes evidence of derivation, dependency, or disciplinary classification.

IX. Scope Constraint

The governing inequality defines structural conditions only.

It does not:

- derive the dissipation function
- specify domain-specific implementations
- introduce additional system variables

X. Canonical Statement

DHFT is defined by invariant structure.

System stability is defined by the governing inequality.

Validity is defined by structural admissibility.

Interpretation does not alter system conditions.

XI. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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